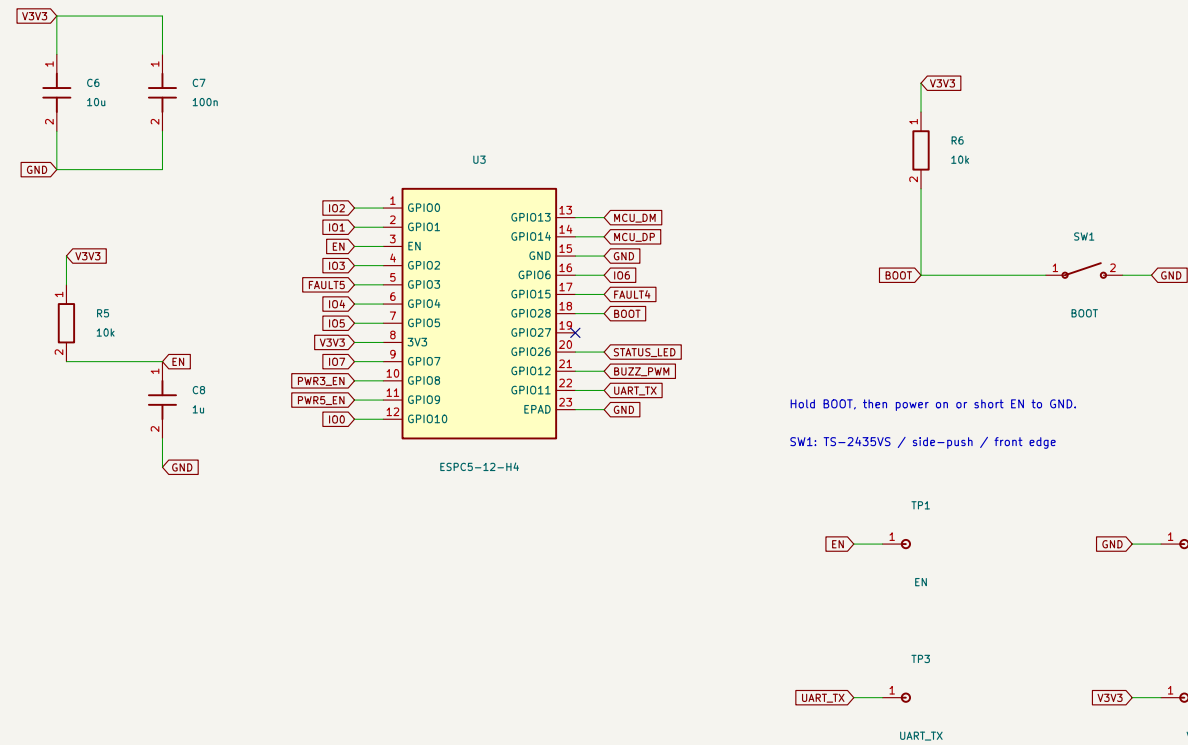
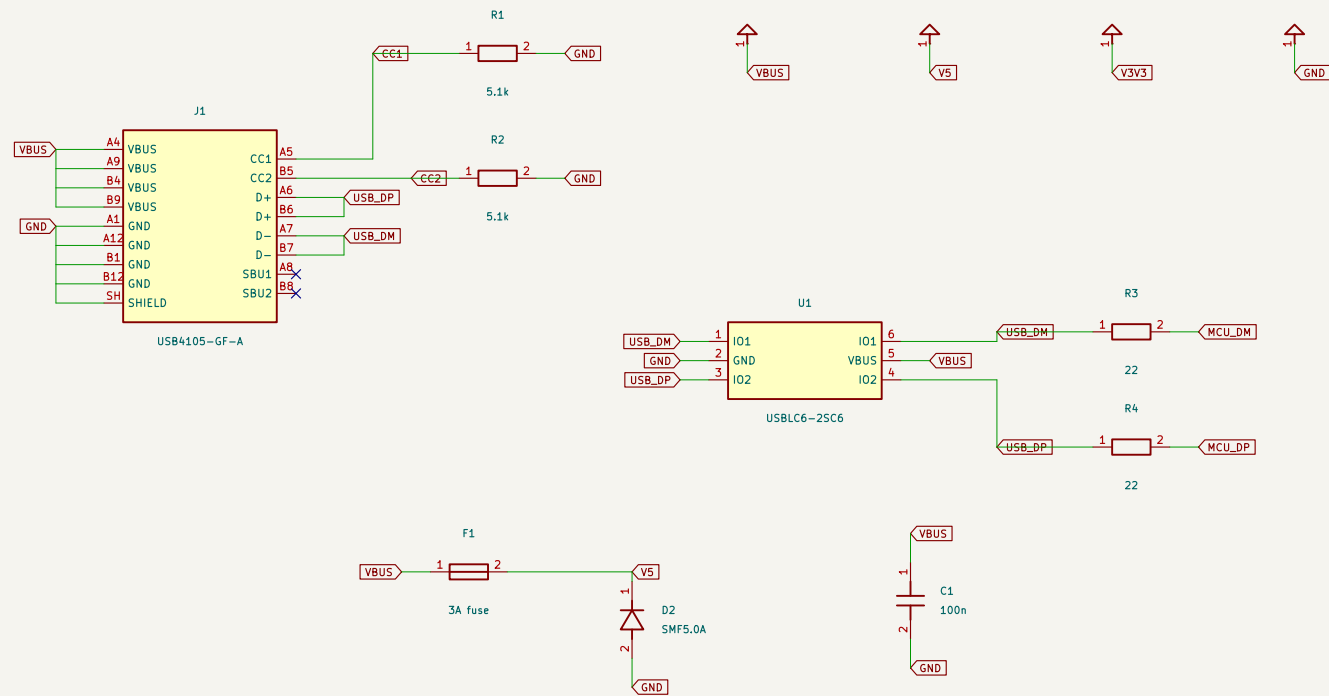


01 CONTROLLER / BOOT / RESET



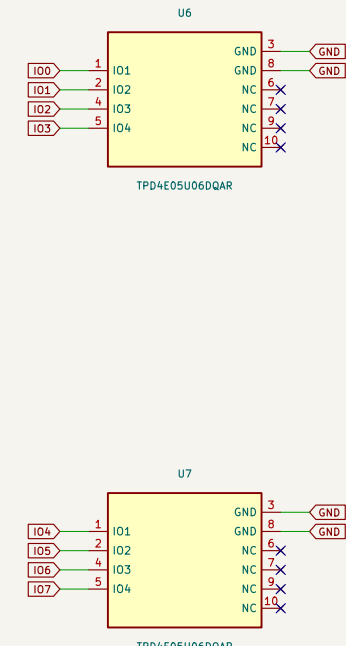
Built-in PCB antenna over cutout; 3.3 V pins only.

02 USB-C / INPUT PROTECTION

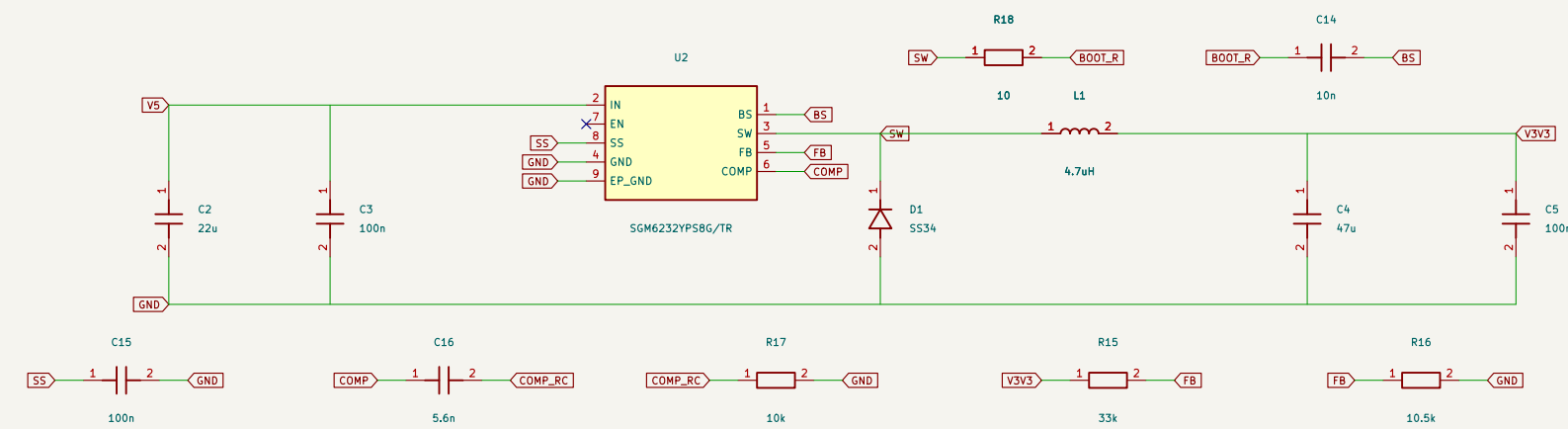


5 V / 3 A supply. No PD or OTG. External loads OFF on PC USB.

07 HEADER ESD / SHUNT PROTECTION

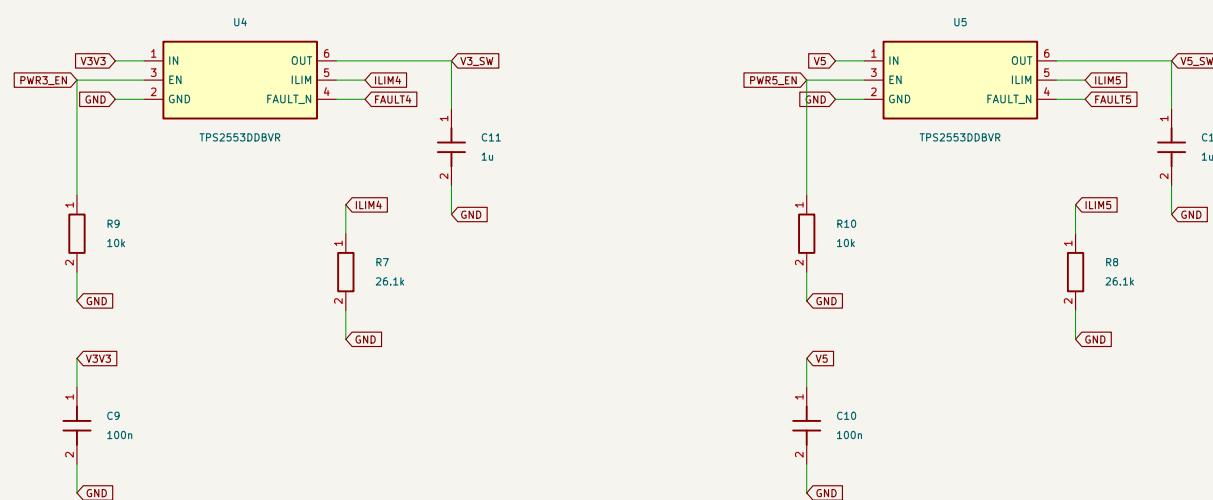


03 SGM6232 / ALWAYS-ON 3.3 V



EN open: auto-start. EP to GND. FB divider $33k / 10.5k = 3.314$ V.

04 SWITCHED OUTPUTS / 1 A NOMINAL LIMIT



Fault inputs: firmware must enable internal pullups.

26.1k ILIM: nominal 0.989 A; bounds 0.908–1.081 A.

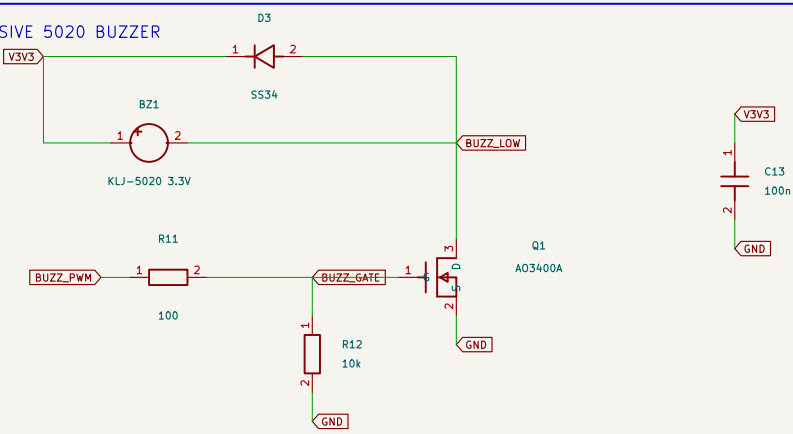
PROTOTYPE VALIDATION

Check startup, dropout, ripple, heat.

Use specified 5 V / 3 A adapter.

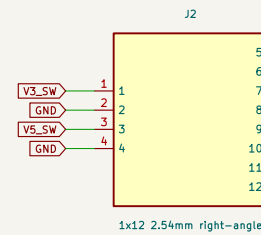
Verify current limit and both loaded rails.

05 PASSIVE 5020 BUZZER



4 kHz / 50% duty. GPIO12 reserved.

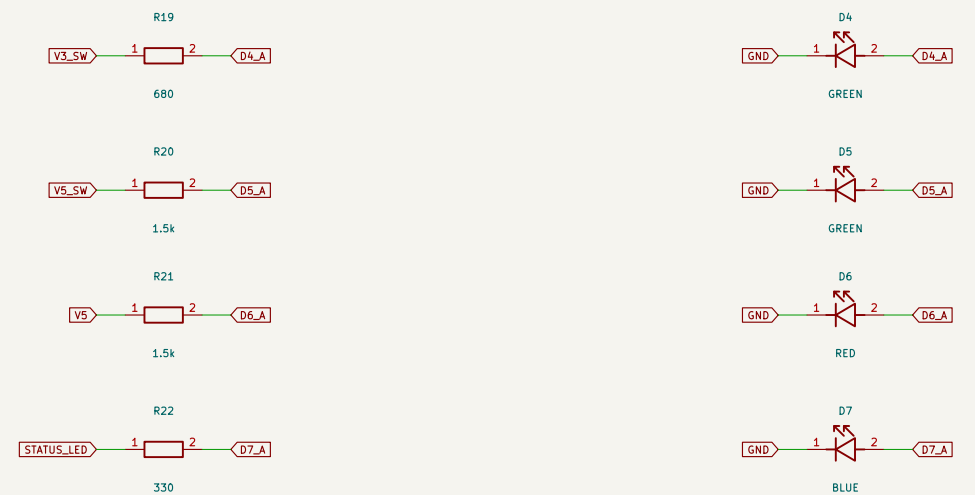
06 12-PIN MALE HEADER



nm right-angle

48 x 28 mm PCB – LEDs top; eight resistors bottom.

08 GROUPED STATUS INDICATORS



Blue: control pin 26, HIGH = on. 3V3/5V: switched rails.

48 x 28 mm / 2 layers / SGM6232 / prototype

GPT6-Astra / xqe2011

Sheet: /

Title: SAIHub-Mini – IO CONTROLLER

Size: A2	Date: 2026-09-24
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KiCad E.D.A. 10.0.6

ev: A

1/1